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Revealing bulk-like properties in freestanding PVD copper foils

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ABSTRACT

The discovery of novel metallic alloys with superior properties for structural applications is needed to address current challenges and enable technological advances. However, high-throughput fabrication of structural materials to explore a broad range of new chemistries is difficult using conventional bulk processing methods. Combinatorial sputter deposition of thin films is commonly used to rapidly explore wide compositional spaces for functional materials but typically produces fine-scale, columnar microstructures that differ from bulk counterparts and remain fixed to a substrate. Here, we demonstrate that a combination of sputter deposition, thermal annealing, and mechanical processing can be used to fabricate thick, free-standing foils with bulk-like microstructures for use in alloy design and discovery. To evaluate this concept, we selected pure copper as a model system, given its well-documented physical and mechanical properties and the extensive prior research available for meaningful comparison. Specifically, we sputter-deposit 200 μm thick Cu foils, remove them from their substrates for thermal and mechanical processing, and show that their mechanical and physical properties closely resemble those of conventionally processed bulk Cu foils.

1. Introduction

The development of metallic alloys with superior properties—such as high mechanical strength, excellent electrical and thermal conductivity, and enhanced oxidation resistance—is essential to meet the ever-increasing performance demands of structural materials in multiple application spaces [1–3]. Consequently, there is a critical need to accelerate the discovery of novel alloys with optimized chemistries and microstructures that can deliver exceptional combinations of physical and mechanical properties. The rapid exploration of alloy composition space has emerged as a promising strategy to address this challenge [4–7].

Conventional alloy development approaches, however, are often too slow and resource-intensive to effectively navigate the vast compositional landscape of multicomponent systems. For example, fabricating even a small subset of possible alloy compositions via traditional methods such as sequential casting or powder metallurgy is impractical, thereby motivating the advancement of high-throughput combinatorial fabrication techniques. Several alternative synthesis strategies—such as additive manufacturing, electrodeposition and physical vapor deposition (PVD)—have been explored to accelerate alloy screening and discovery [8–11].

Additive manufacturing enables the fabrication of compositionally graded structures and near-net-shape components, providing a viable route to assess new alloys in near-bulk form [12]. However, challenges such as porosity, anisotropic microstructures, and residual stress can limit its effectiveness for certain compositions and applications [13]. Electrodeposition can deposit thick metallic foils at high rates and low-cost and recently has been explored as a combinatorial or high-throughput approach in electrocatalysis, battery materials, and novel nanomaterials discovery [14–16]. However, scalability and instrument-related constraints (e.g., current distribution, mass transport, and cell design) remain and can hinder film uniformity and substrate size [17]. In contrast, PVD methods, particularly magnetron sputtering, have gained significant attention as combinatorial tools due to their ability to co-deposit multiple elements with precise compositional control [18,19]. This technique enables the creation of composition gradients or discrete libraries on a single substrate, allowing rapid

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evaluation of composition–property relationships across a broad chemical space [20,21]. Sputtering has proven especially useful for the discovery of functional alloys, significantly reducing the time required to identify promising candidate systems [22,23].

Despite these advantages, sputter-deposited thin films introduce inherent challenges when evaluating bulk structural properties. These films are typically limited to micrometer or sub-micrometer thicknesses and often exhibit columnar, fine-grained microstructures that differ markedly from those observed in bulk-processed materials [24,25]. It is well established that film thickness can strongly influence mechanical properties; nanoscale grain structures and dimensional constraints in films tend to elevate hardness or strength relative to their bulk counterparts [26,27]. Moreover, these films are typically substrate-bound, which complicates thermal and mechanical processing, and inhibits direct measurement of intrinsic properties such as tensile strength, fracture toughness, or creep behavior. Substrate effects can also influence grain growth and phase evolution during annealing, further confounding attempts to extrapolate thin-film behavior to bulk conditions [28,29].

To overcome these limitations, the fabrication of freestanding metallic foils of sufficient thickness via sputtering offers a promising solution. By decoupling the film from its substrate and increasing its thickness to the scale of bulk specimens (tens to hundreds of microns), the material can be processed, handled, and tested using conventional techniques [30]. Thick multilayer foils have been synthesized to investigate self-propagating exothermic reactions in various Al/Ni-based systems [31–33]. These freestanding foils permit heat treatments and mechanical testing without substrate interference, enabling grain growth and phase evolution similar to those observed in cast or wrought materials. For structural studies, 100 μm-thick Cu/Nb microlaminates were fabricated to study creep deformation in fine-grained niobium at elevated temperatures (~600 °C) [34,35]. Similarly, ~30 μm-thick nanotwinned Ni-Mo-W foils were sputter deposited and exhibited excellent thermomechanical stability and favorable electrical properties for MEMS applications [36,37]. In another study, microlaminated Mo/W foils (~50 μm thick) were synthesized to independently control strength and toughness by tuning layer thickness [38]. This approach of fabricating thick, free-standing foils by sputtering has the potential to bridge the gap between high-throughput thin-film synthesis and conventional bulk alloy development, thereby enabling meaningful and representative structure–property correlations.

In this study, we present a novel, proof-of-concept demonstration of bulk-like behavior in free-standing, sputter-deposited metallic foils by comparing their microstructures and properties to those obtained from chemically equivalent foils that were processed by traditional bulk processing methods. Pure copper was selected as the model system due to its well-established physical and mechanical properties and the substantial amount of prior research available [39–42]. Approximately 200 μm thick foils were fabricated via direct current magnetron sputtering and mechanically released from their substrates. Following thermal and mechanical processing, the microstructure and physical properties of the PVD foils were evaluated and compared to those of commercially and widely available cast and rolled (C/R) Cu foils of similar thickness and chemistry. Our results show that the sputtered, released, and processed foils can achieve comparable bulk-like microstructures as well as mechanical and physical properties. This work opens a new pathway for high-throughput alloy discovery by enabling direct assessment of bulk properties in combinatorially synthesized materials using thick foil PVD processing.

2. Experimental methods

2.1. Fabrication

Approximately 200 μm thick Cu foils were fabricated via direct current (DC) magnetron sputtering using a custom-designed PVD system, as shown in Fig. 1. Foils were deposited onto mirror-finish polished brass substrates mounted on a rotating, water-cooled octagonal carousel using three 305 mm × 127 mm linear Cu targets (99.99 % OFHC Cu, Process Materials Inc.). The targets were symmetrically arranged to ensure uniform coating. Sputtering was performed at 850 W and 1.8 mTorr of argon pressure. The deposition rate from each Cu target was approximately 1 μm/h. Prior to deposition, the chamber was evacuated to a base pressure of $\sim 5 \times 10^{-6}$ Torr. To fabricate freestanding foils, the substrate was patterned with tape to delineate 1.5″ × 1.5″ squares, allowing easy detachment via mechanical flexure after deposition. For comparative analysis, 250 μm thick C/R Cu foils (99.99 % purity, oxygen-free high conductivity) were obtained from Goodfellow Advanced Materials.

2.2. Thermal and mechanical processing

Both the PVD and C/R Cu specimens were subjected to annealing at 300 °C, 500 °C, and 700 °C for 4 h in a horizontal quartz tube furnace under a reducing atmosphere consisting of 96 % N_2 and 4 % H_2. To minimize residual oxygen, the furnace was initially evacuated to $\sim 3 \times 10^{-3}$ Torr and then purged with the reducing gas mixture; this evacuation–purge cycle was repeated twice before heating.

The foils were placed on clean, flat alumina boats to avoid contamination and ensure uniform heat exposure. A calibrated thermocouple was positioned adjacent to the specimens to accurately monitor temperature near the specimen surface. Heating was performed at a ramp rate of 20 °C/min. After reaching the target temperature, the specimens were held isothermally for 4 h, followed by furnace cooling to room temperature under the same reducing atmosphere.

To demonstrate the ability to mechanically process PVD foils, one annealed sample was rolled to a 50% thickness reduction in a single pass at room temperature.

2.3. Microstructural characterization

Microstructural evaluation was performed using electron backscatter diffraction (EBSD). Specimens were initially ground using SiC papers of progressively finer grit sizes (up to 4000 grit), followed by cloth polishing with colloidal silica suspension to achieve a mirror finish. Final surface preparation was carried out by ion milling using a Leica EM TIC 3X at 5 kV and 3 mA for 10 min to remove any surface deformation layers. EBSD scans were performed using a Tescan MIRA 3 GM field emission scanning electron microscope (FE-SEM) equipped with a Symmetry, Oxford Instruments EBSD detector, operated at 20 kV accelerating voltage and 15 mA beam current. Orientation maps were acquired with a step size of 0.01–0.5 μm, depending on the grain size of the specimen.

Grain size distribution, crystallographic orientation, and texture components were analyzed using Oxford AZtecCrystal software and MTEX [43]. The grain boundaries were classified using a 15° misorientation threshold for high-angle grain boundaries; Σ3 boundaries were identified using a Brandon criterion (60° ± 5° about ⟨111⟩-axis) and were counted as grain boundaries in the statistics. Grain sizes were determined from EBSD orientation maps using the equivalent circle diameter approach, where each indexed grain is represented by a circle of equivalent area. The reported values correspond to the number-weighted (arithmetic) mean of all measured grain diameters, with standard deviations calculated from the same distributions. This approach was applied consistently across all specimens to ensure comparability and is appropriate for Hall-Petch (H–P) analysis because it avoids overemphasizing a small number of abnormally large grains and underrepresenting the fine-grain population.

2.4. Mechanical property measurements

a Microhardness testing

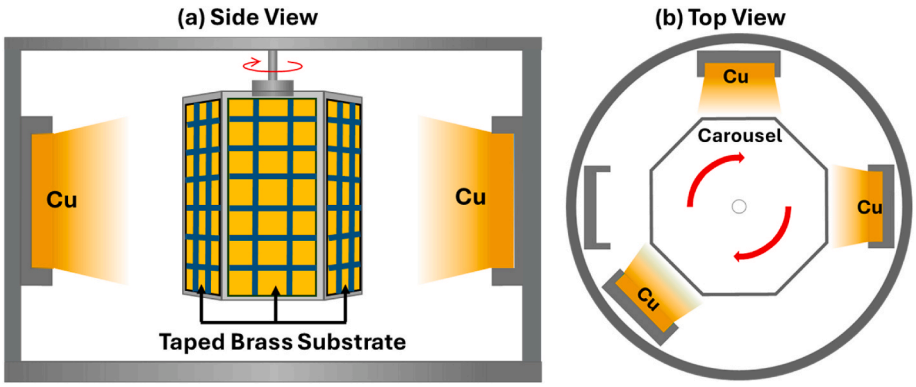


Fig. 1. (a) Side and (b) top view schematic of deposition chamber configured for Cu deposition on brass substrates with tape to separate regions into 1.5in x 1.5in foils.

Vickers microhardness measurements were performed using a LECO AMH55 microhardness tester with a load of 50 gf and a dwell time of 15 s. At least ten indentations were made per specimen to ensure statistical reliability. Indents were spaced at a distance of at least six times the diagonal length of the impression to prevent plastic zone interaction. Hardness values were calculated automatically using the instrument's Cornerstone software, and average values with standard deviations were reported.

b Tensile testing

Dog-bone–shaped tensile specimens were fabricated by wire electrical discharge machining (EDM). The gauge length and width were 5.33 mm and 1.00 mm, respectively. Each specimen was lightly polished on both sides to minimize surface defects and ensure uniform strain distribution. Uniaxial tensile tests were conducted using an MTS Criterion C43 universal testing machine at a nominal engineering strain rate of $\sim 1 \times 10^{-3}\ s^{-1}$. Stress was calculated by dividing the applied load by the initial cross-sectional area of the gauge section. Strain was determined based on crosshead displacement normalized by the gauge length. The resulting engineering stress–strain curves were used to evaluate yield strength, tensile strength, and uniform elongation. The yield strength was determined using the 0.2 % offset method.

2.5. Thermal analysis

Thermal behavior, including melting point and oxidation resistance, was evaluated using a TA Instruments SDT650 simultaneous differential thermal analysis/thermogravimetric analysis (DTA/TGA) system. Each specimen had a mass of approximately 15 mg and was placed inside a clean alumina crucible for the measurements. DTA experiments were conducted under flowing ultra-high-purity argon (>99.999 %) at a flow rate of 100 mL/min. Prior to heating, the system was purged with argon for 10 min to remove residual oxygen and moisture. The specimens were heated from room temperature to 1150 °C at a rate of 20 °C/min. The melting point was determined using the tangent method on the DTA curve, defined as the intersection between the extrapolated baseline and the leading edge of the endothermic melting peak. To evaluate oxidation behavior, TGA scans were performed by heating the specimen from room temperature to 1000 °C at 20 °C/min in air at a flow rate of 100 mL/min.

2.6. Electrical resistivity measurements

Room-temperature electrical resistivity was measured using a standard four-point probe setup. The specimens were mounted flat on a clean glass substrate to ensure a uniform and stable contact surface. Four equally spaced metal probes were gently brought into contact with the top surface of the foil, ensuring consistent probe spacing and minimal indentation.

Current was applied through the outer probes in the range of −50 mA to +50 mA in 5 mA increments using a source meter. The voltage drop across the inner probes was recorded for each current step. Electrical resistance was extracted from the slope of the linear fit to the current–voltage (I–V) data. Resistivity (ρ) was then calculated using the standard geometric correction factors based on specimen thickness and probe spacing. Each measurement was repeated at five different locations per specimen to account for possible local variation, and the average resistivity was reported.

3. Results

3.1. Representative specimens of as-deposited PVD Cu foils

Fig. 2a shows the as-deposited PVD foil on a 5″ x 12″ brass substrate separated into individual square dimensions of 1.5" highlighted in a red dashed box. A total of 18 such squares are generated on each substrate and 8 such substrates can be used during a single deposition. This demonstrates the high-throughput capability yielding up to 144 specimens per deposition when eight substrates are processed simultaneously. Fig. 2b displays a freestanding foil removed from the substrate; samples such as these were used to generate specimens for all experiments. Fig. 2c presents dog-bone-shaped specimens machined via wire EDM, which were subsequently employed in uniaxial tensile testing to assess mechanical performance.

3.2. Thermal processing and microstructural evolution

Fig. 3a–d presents orientation maps and corresponding {100}, {110}, and {111} pole figures for the PVD Cu specimens in as-deposited and annealed conditions, illustrating the combined effects of thermal treatment on both the grain structure and crystallographic texture evolution. The as-deposited foil (Fig. 3a) exhibits a nanocrystalline grain structure with an average grain size of approximately 100 nm.

Annealing at 300 °C (Fig. 3b), does not significantly alter the grain size, indicating limited grain boundary mobility at this temperature. Interestingly, early signs of abnormal grain growth are observed as highlighted in dashed enclosed region in Fig. 3b, suggesting the onset of grain boundary migration in a few energetically favorable orientations. The Σ3 twin boundary fraction shows a modest increase from 38.0 % to 41.2 % (Table 1), reflecting initial boundary relaxation and minor twin formation.

Upon annealing at 500 °C (Fig. 3c), abnormal grain growth becomes prominent, resulting in a bimodal grain size distribution comprising of large grains (~8 μm) embedded within a matrix of finer grains (~200 nm). The grain size increase combined with copper's relatively low

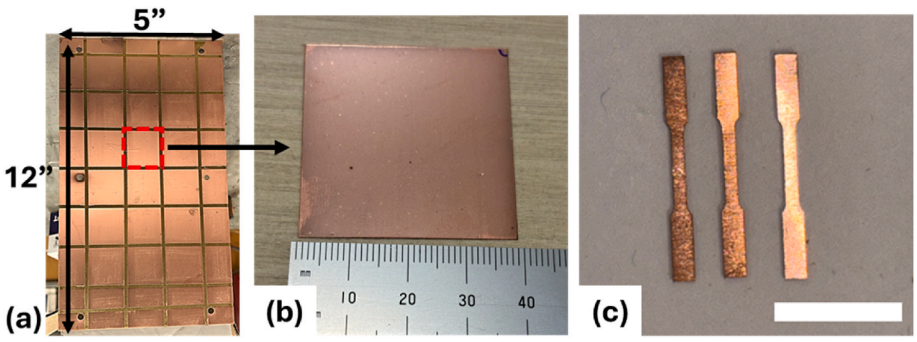


Fig. 2. Representative picture of PVD Cu (a) deposited on a 5″ x 12″ brass substrate (b) freestanding foil removed from the brass substrate (c) dog-bone shape fabricated by wire EDM for tensile testing. The scale bar in (c) corresponds to 1 cm.

stacking fault energy (SFE), promotes the nucleation of annealing twins [44]. The Σ3 fraction increases substantially to 52.8 %, indicating that twin-assisted recrystallization is becoming a dominant mechanism at this stage. These annealing twins have a preferred (100) orientation to minimize the strain energy in the newly formed grains. Further annealing at 700 °C leads to further grain growth (average grain size ~ 2.1 μm) and an increase in the fraction of annealing twins, as evidenced by a high Σ3 boundary fraction of 65.9 %.

The pole figures in Fig. 3 clearly demonstrate the textural evolution as PVD specimens are annealed. The as-deposited state showed a strong {111} texture that shifted gradually towards ⟨100⟩ with increasing temperature. When these foils are annealed at higher temperatures (~1000 °C) for longer periods, a dominant {100} texture has been observed as shown in Supplementary Fig. S1.

Fig. 4a–d presents orientation maps and corresponding {100}, {110}, and {111} pole figures of the C/R Cu specimens in as-received and annealed conditions. The as-received rolled Cu foil (Fig. 4a), exhibits a typical deformation texture characterized by a mixture of brass-type {011}⟨211⟩, copper-type {112}⟨111⟩, and S-type {123}⟨634⟩ texture components, with an average grain size of approximately 2.5 μm [45].

Upon annealing at 300 °C (Fig. 4b), the foil undergoes recovery, likely through dislocation annihilation and the formation of dislocation walls as reported in literature [46–48]. This results in a slight weakening of the existing deformation texture without significant changes in the dominant orientation components. The average grain size increases to 4.5 μm, and the Σ3 twin boundary fraction increases from 44.5 % to 64.1 % (Table 1), indicating further increase of annealing twinning.

After annealing at 500 °C (Fig. 4c), recrystallization becomes more pronounced, grain size increases to ~8.3 μm, and the pole figures reveal a pronounced weakening of the initial deformation texture. The Σ3 twin boundary fraction remains high at 61.6 %, reflecting active twin formation/growth during grain growth and boundary migration, common in materials like Cu with relatively low SFE [44].

Upon annealing at 700 °C (Fig. 4d), the texture observed in the as-received condition remains discernible, although noticeably weaker. As shown in Table 1, the average grain size increases substantially from 2.5 μm in the as-received state to approximately 32 μm after annealing. However, the Σ3 twin boundary fraction drops sharply to 16.2 %.

To assess the grain morphology through the foil thickness, cross-sectional EBSD scans were performed on PVD specimens annealed at 300 °C and 700 °C, as shown in Fig. 5a, b. The micrographs reveal multiple grains across the foil thickness. This grain structure is likely due to the discontinuous deposition process inherent to the rotating carousel configuration in the deposition chamber. As the carousel rotates during deposition, periodic shadowing leads to localized interruptions in film growth, thereby promoting grain boundary formation and fragmentation of growing columns. This behavior is analogous to pulsed deposition techniques, where off-times in the deposition cycle promote surface diffusion and facilitate the formation of equiaxed grains instead of continuous columnar structures [49,50].

At 300 °C (Fig. 5a), the microstructure exhibits fine equiaxed grains distributed uniformly through the thickness. There is little evidence of abnormal growth consistent with earlier plan-view EBSD observations. After annealing at 700 °C (Fig. 5b), multiple grains are evident across the film thickness, with well-developed annealing twins within the larger grains. These twins appear as lamellar features within larger grains, consistent with the low stacking fault energy of Cu, which promotes twin formation during grain boundary migration and recrystallization. The presence of multiple large, twinned grains through the thickness also confirms that recrystallization has occurred throughout the specimen.

3.3. Mechanical processing of PVD Cu

To demonstrate that mechanical processing can be performed on PVD specimens, a PVD Cu foil annealed at 700 °C was cold rolled to achieve a 50 % thickness reduction in a single pass. The imposed mechanical deformation led to dynamic recrystallization-like behavior, resulting in finer grains with an average size of approximately 1.1 μm. The finer grain size is attributed to dynamic recrystallization as reported by others following room temperature deformation of copper [51,52]. Texture analysis via pole figures reveals a dominant brass-type texture (Fig. 6a), which is typically observed in rolled Cu [41,53]. The diffuse nature of the poles suggests the presence of local intragranular misorientation that may have evolved during the deformation.

To observe the microstructural evolution following subsequent thermal processing, the cold-rolled foil was annealed at 500 °C for 4 h. EBSD analysis reveals a uniform but vastly larger grain structure with an average grain size of ~26.2 μm (Fig. 6b), which contrasts significantly with the microstructure of an as-deposited PVD Cu annealed at the same temperature. Unlike the latter, which exhibited a bimodal grain distribution due to abnormal grain growth, the rolled and annealed PVD specimen displayed a more homogeneous grain structure. This difference is attributed to thermomechanical processing, which likely suppresses the development of abnormal grains through texture evolution. The pole figures show a dominant R texture which has been observed to develop through selective boundary migration mechanisms during annealing in brass-textured microstructures [54,55]. These results clearly demonstrate the ability to induce large plastic strains in PVD foils through conventional mechanical processing techniques to effectively tailor their microstructure, both in terms of grain size and crystallographic texture, much like bulk Cu.

3.4. Tensile properties and hardness

In Fig. 7a, the PVD specimens display a wide range of stress–strain curves that vary with annealing condition. The as-deposited foil exhibited a typical response for nanocrystalline Cu, with a high yield strength of ~335 MPa and minimal ductility, consistent with literature

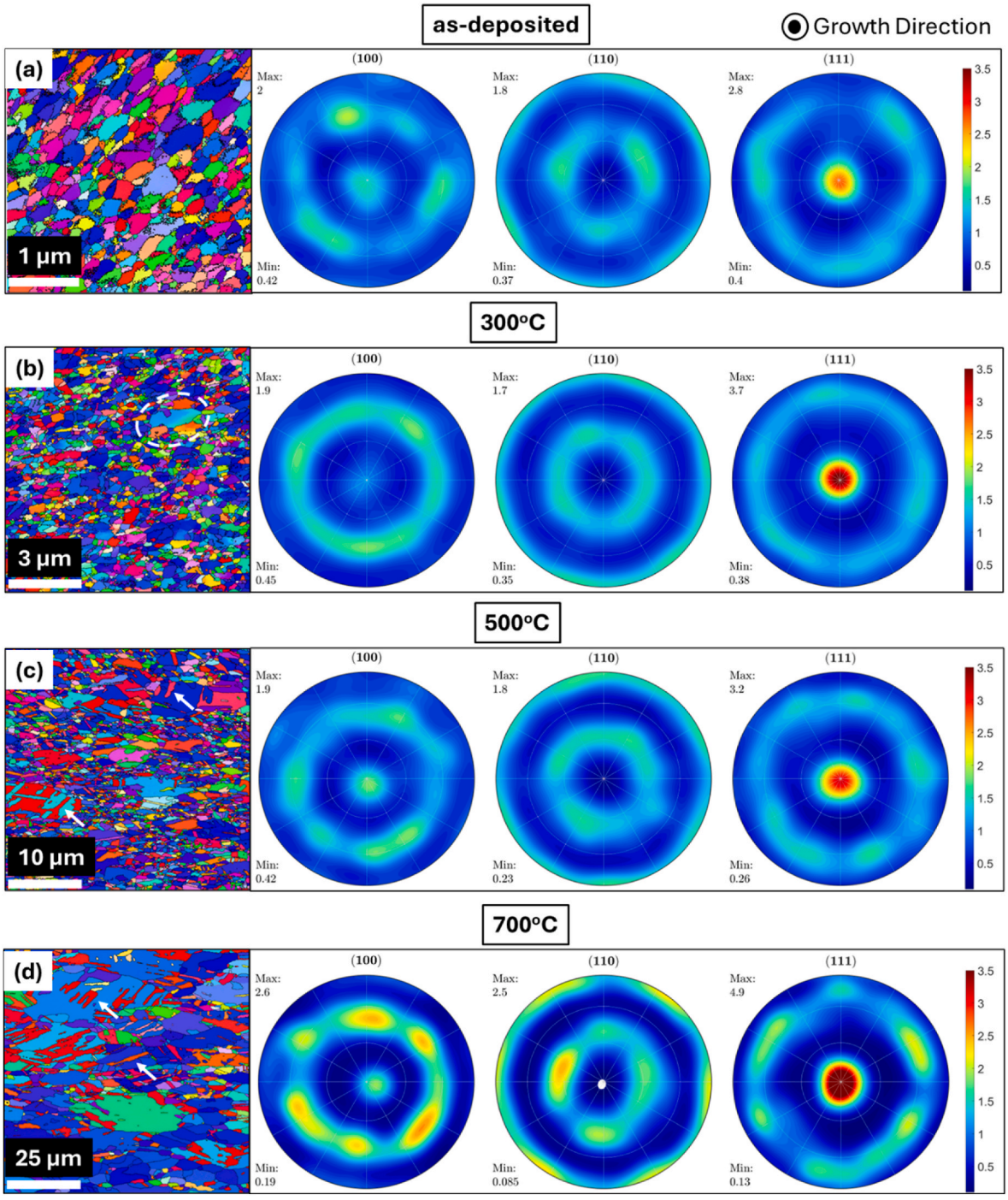


Fig. 3. EBSD inverse pole figure (IPF) orientation maps and corresponding {100}, {110}, and {111} pole figures of PVD Cu specimens in (a) the as-deposited condition and after annealing at (b) 300 °C, (c) 500 °C, and (d) 700 °C for 4 h. The dashed enclosed region in (b) highlights an abnormally large grain relative to the surrounding matrix, indicative of the onset of abnormal grain growth. Arrows in (c) and (d) denote the formation of annealing twins at elevated temperatures. All scans were obtained from the plan-view surface. A consistent intensity scale is used across all pole figures to enable direct comparison of texture evolution with annealing temperature.

reports on nanocrystalline Cu specimens [39]. The negligible strain hardening observed is attributed to limited dislocation storage, interaction, and multiplication due to the presence of smaller grains, resulting in early strain localization and fracture.

After annealing at 300 °C, the yield strength decreased to ~270 MPa, while the ductility increased to 17 %. Although strain hardening remained minimal, strain localization was delayed, likely due to recovery processes in the microstructure and the onset of abnormal grain growth. Further annealing at higher temperatures led to a continued decrease in strength and an increase in ductility. The occurrence of abnormal grain growth resulted in a bimodal grain structure, which contributed to a modest increase in strain hardening rate compared to as-deposited and 300 °C annealed conditions.

Fig. 7b shows the engineering stress–strain behavior of C/R Cu specimens. The as-received specimen exhibited a 0.2 % offset yield strength of approximately 128 MPa, which decreased to 65 MPa after annealing at 700 °C. All specimens were able to elongate uniformly up to 45 % strain, accompanied by a continuous decrease in the strain

Table 1
Σ3 twin boundary fraction and average grain size of PVD and C/R Cu in the as-deposited/as-received condition and after annealing at 300 °C, 500 °C, and 700 °C for 4 h.

	initial	300 °C	500 °C	700 °C
Σ3 fraction (PVD Cu)	38.0	41.2	52.8	65.9
Σ3 fraction (C/R Cu)	44.5	64.1	61.6	16.2
Grain size (μm) (PVD Cu)	0.1 ± 0.06	0.2 ± 0.1	0.4 ± 0.3	2.1 ± 2
Grain size (μm) (C/R Cu)	2.5 ± 2	4.5 ± 3.9	8.3 ± 6.4	32 ± 24.5

hardening rate. The mechanical response of these specimens is primarily governed by grain boundaries and dislocation strengthening mechanisms, where dislocations either multiply within grains or are impeded by grain boundaries during their motion. These specimens showed a typical strain hardening behavior exhibiting a higher hardening rate just after yielding and gradual reduction with strain.

Since PVD and C/R Cu exhibit different crystallographic textures, the tensile yield stress reflects not only microstructural features (grain size, twins, dislocation density) but also texture effects (via Taylor factors and latent hardening). As a result, Hall–Petch (H-P) constants derived from uniaxial tension can appear texture dependent for PVD vs C/R Cu. To

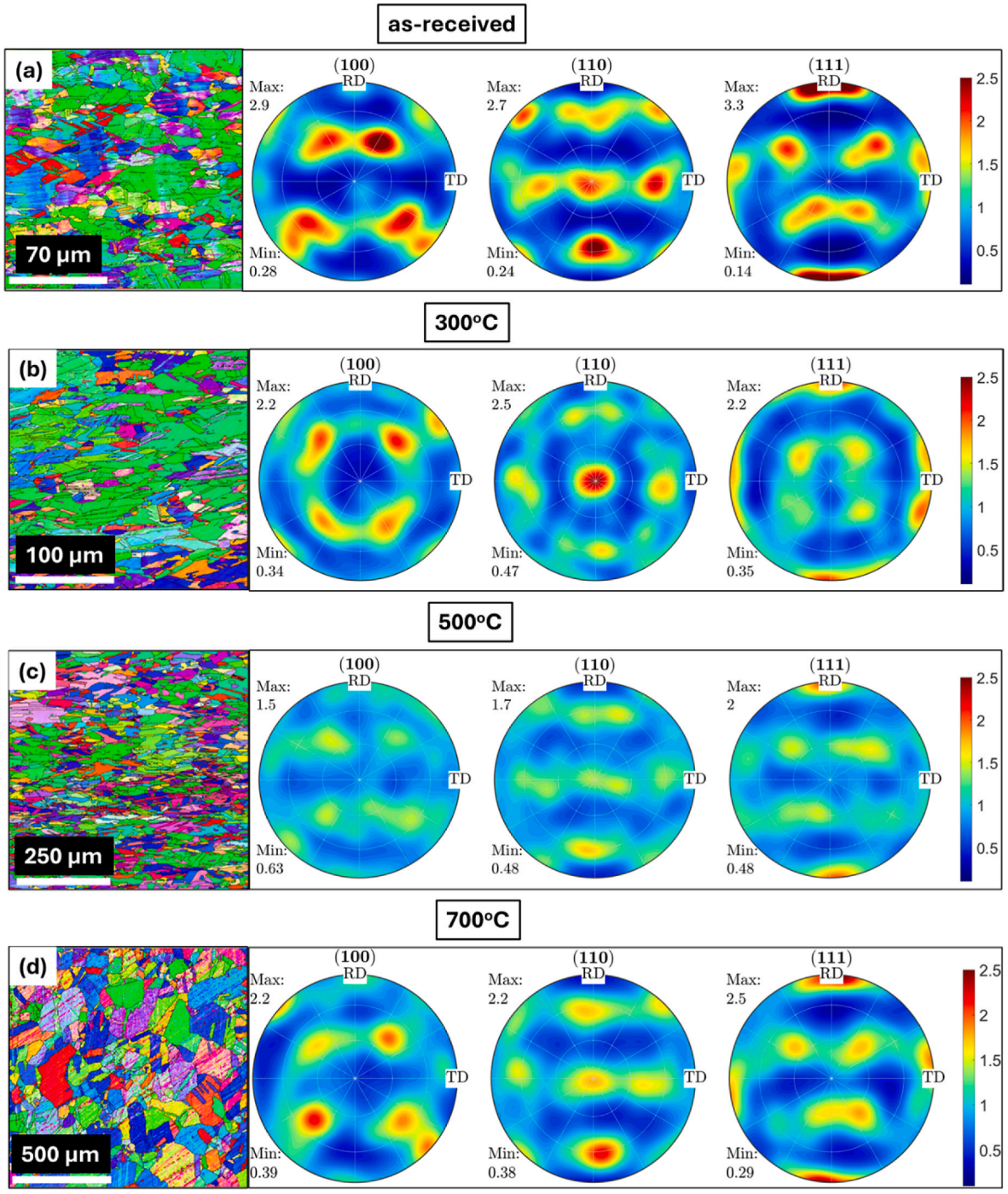


Fig. 4. (a–d) EBSD inverse pole figure (IPF) orientation maps and corresponding {100}, {110}, and {111} pole figures of C/R Cu specimens in (a) the as-received condition and after annealing at (b) 300 °C, (c) 500 °C, and (d) 700 °C for 4 h. All scans are captured from the plan-view surface, with a uniform intensity scale applied to all pole figures for accurate comparison of texture evolution.

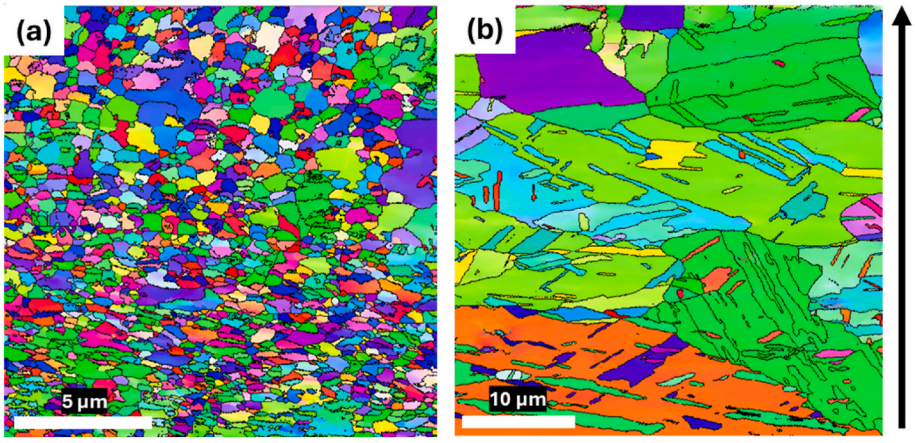


Fig. 5. Cross-sectional EBSD inverse pole figure (IPF) orientation maps of PVD Cu annealed at (a) 300 °C and (b) 700 °C for 4 h. The micrographs reveal the through-thickness grain morphology, indicating the development of equiaxed grains and annealing twins at elevated temperatures. The black arrow in the figure represents the Cu growth direction during deposition.

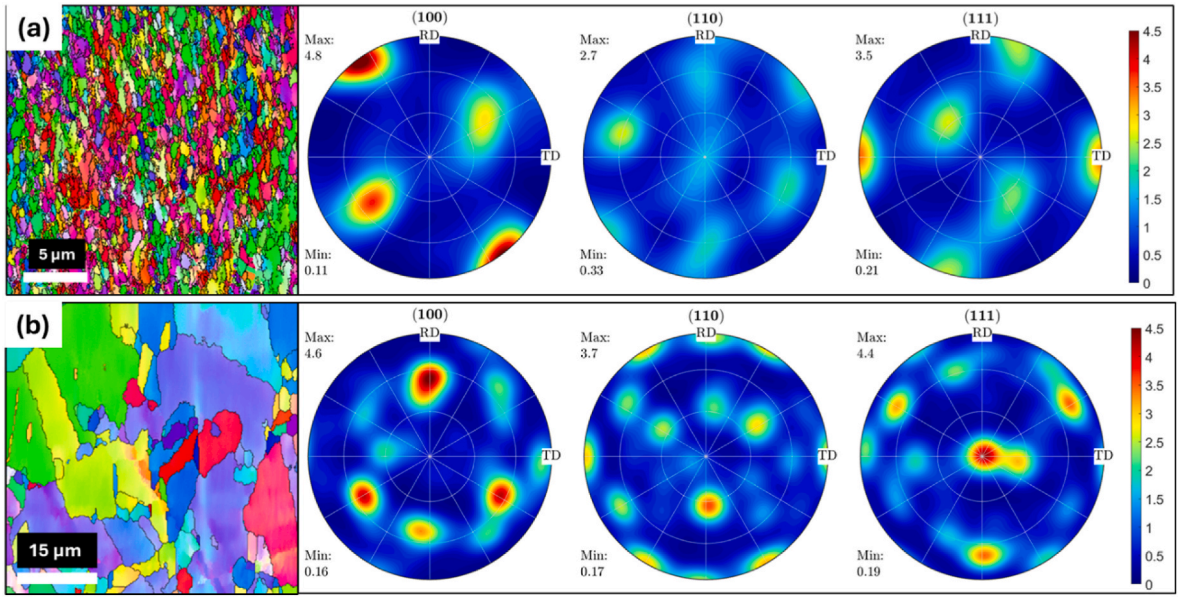


Fig. 6. Plan-view EBSD inverse pole figure (IPF) orientation maps and corresponding {100}, {110}, and {111} pole figures of 700 °C annealed PVD Cu specimen after (a) 50 % cold rolling and (b) subsequent annealing at 500 °C for 4 h. The maps demonstrate the development of brass-type deformation texture after rolling and R-texture following recrystallization.

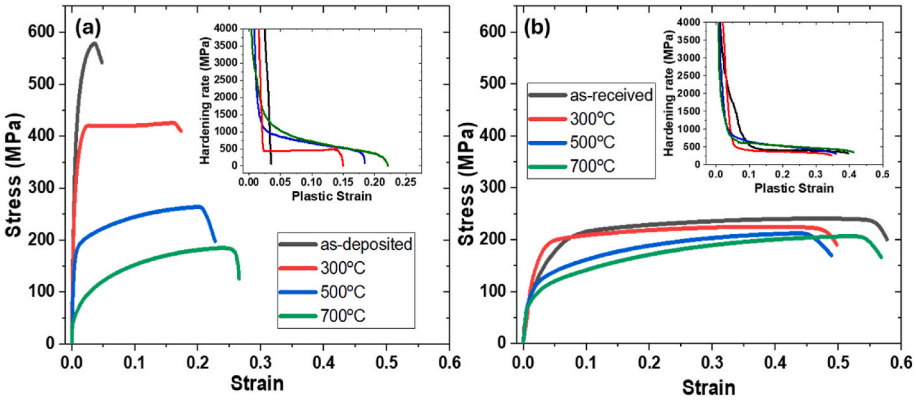


Fig. 7. Engineering stress–strain curves of (a) PVD and (b) C/R Cu specimens in their initial condition and after annealing at 300 °C, 500 °C, and 700 °C. Inset shows the strain hardening rate of each specimen at different conditions.

minimize texture bias when establishing H–P relationships, Vickers indentation was utilized in addition to tension testing. The multiaxial, confined plasticity under the indenter activates multiple slip systems and averages over many grain orientations within the plastic zone, producing high local strains while reducing sensitivity to macroscopic texture [56].

Fig. 8 illustrates the variation in hardness of both PVD and rolled Cu as a function of annealing temperature. For the PVD Cu, a pronounced decrease in hardness is observed from approximately 216 HV in the as-deposited condition to 74 HV after annealing at 700 °C. In contrast, the rolled foils exhibit a more gradual reduction, with hardness decreasing from 92 HV in the as-received state to 61 HV after annealing at 700 °C. This trend aligns with the inverse square-root relationship between hardness and grain size. For the PVD Cu, the grain size increases from ~0.1 μm to ~2.1 μm, corresponding to a $d^{-1/2}$ change from 3.2 to 0.7. Meanwhile, the rolled foils show a grain size increase from ~2.5 μm to ~32 μm, with a $d^{-1/2}$ change from 0.6 to 0.18. The larger change in the PVD specimens ($\Delta d^{-1/2} \approx 2.5$) compared to the C/R specimens ($\Delta d^{-1/2} \approx 0.42$) explains the more significant hardness drop, as also reflected in the H–P plot in Fig. 9a.

Fig. 9 presents the variation of hardness and yield strength as a function of grain size, along with linear fits to the H–P relationship for both foil types. Both measurements reveal similar H–P trends within each foil type. Notably, the PVD Cu exhibit a slightly lower H–P slope and lower intercept compared to the C/R Cu, indicating a reduced grain boundary strengthening effect. The differences in slope and intercept observed suggest that other deformations mechanisms could be contributing to the overall strength. Given that all specimens are high-purity Cu, the influence of solid solution or particle strengthening on the intercept can be ruled out.

The as-received, C/R specimens exhibit additional strengthening due to Taylor hardening, leading to higher hardness and yield strength. As a result, when the as-received data points are excluded from the H–P fitting, the resulting slopes for the C/R Cu are lower. When recalculated using only the annealed C/R specimens, a slope of 126 MPa $\mu m^{-1/2}$ (or 59 HV·$\mu m^{-1/2}$) is obtained which is closer to that of the PVD specimens, yet still marginally higher. The underlying factors contributing to the differences in H–P constants between C/R and PVD Cu are elaborated in the discussion section. Furthermore, when PVD foils were subjected to bulk mechanical processing (Section 3.3), the influence of microstructural evolution was assessed through hardness testing. The measured hardness values aligned well within the linear H–P trends established for both PVD and C/R Cu.

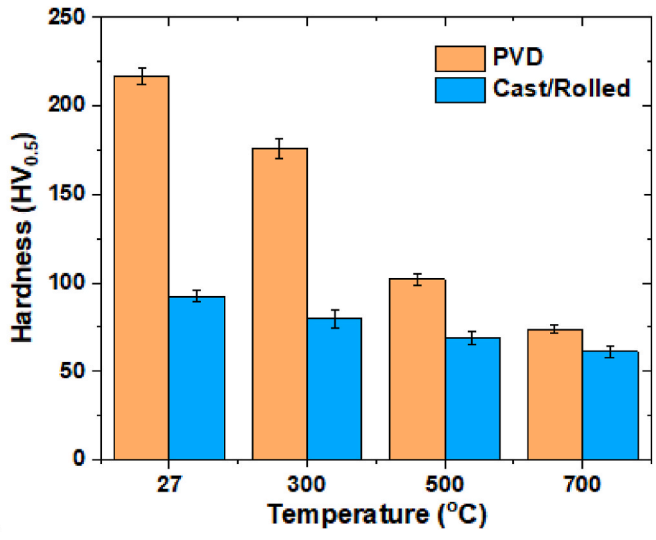


Fig. 8. Vickers microhardness of PVD and C/R Cu as a function of annealing temperature.

3.5. Thermal properties and oxidation behavior

Fig. 10 shows DTA scans of as-received C/R, as-deposited PVD, and 700 °C annealed PVD Cu, measuring heat flow as a function of temperature. All scans have a distinct endothermic peak for the melting of Cu. The melting point for as-deposited PVD Cu was 1076 °C, which was quite comparable to as-received C/R Cu (1078 °C). A minor difference of 2 °C is likely attributed to the finer grain size in the as-deposited PVD Cu. This was validated by performing a DTA run on a 700 °C annealed PVD Cu that showed a similar grain size to the as-received C/R Cu. The melting point matched with as-received C/R Cu, demonstrating the similar behavior between PVD and C/R specimen. Simultaneous TGA scans were performed in argon on both material types by heating them up to 1150 °C. The mass gain was observed to be well within 1 %, which is expected as the scan was performed in an inert atmosphere.

To evaluate the oxidation behavior, TGA scans were conducted in air on as-received C/R Cu, as-deposited PVD Cu, and PVD Cu annealed at 700 °C. All three specimens exhibited broadly similar oxidation trends, with some distinctions in the onset and progression of oxidation. For the PVD specimens, the as-deposited foil began oxidizing at ~250 °C, while the annealed foil showed a delayed onset. The percentage mass-gain curves for the two PVD foils converged near ~750 °C.

When comparing annealed PVD Cu to C/R Cu, the oxidation behavior remained similar up to ~600 °C, which is consistent with their comparable grain sizes and the relatively slow oxidation kinetics at lower temperatures. Beyond 600 °C, the oxidation rate increases significantly, leading to rapid mass gain. As shown in Fig. 11, the annealed PVD Cu exhibited higher percentage mass gain (5.14 %) compared to C/R Cu (4.14 %). While this may suggest reduced oxidation resistance in PVD Cu, it is essential to account for geometric effects associated with specimen thickness. The PVD specimens were approximately 200 μm thick, whereas the C/R specimens were 250 μm thick. Given the comparable surface area, a thinner specimen has lower initial mass, thereby yielding a higher percentage mass gain for the same amount of oxide formation. A 20 % decrease in thickness is expected to correspond to a ~25 % increase in percentage mass gain, which aligns well with the experimentally observed difference. Although the PVD and C/R Cu exhibit differences in oxidation onset and progression due to grain size, their overall oxidation behavior remains comparable in terms of total mass gain when normalized for thickness.

3.6. Resistivity measurement

Fig. 12 presents a comparison of the electrical resistivity of PVD and C/R Cu specimens in both as-prepared and annealed states. For the PVD Cu, resistivity decreases progressively with increasing annealing temperature and scattering is reduced. The as-deposited PVD specimen exhibits the highest resistivity value of 2.4 μΩ cm, attributed to significant grain boundary scattering due to its ultrafine grains [40]. After annealing at 300 °C and 500 °C, resistivity declines but remains above the bulk value. After 700 °C, the microstructure is more homogeneous with larger grains, and the resistivity approaches the annealed Cu value. In contrast, the C/R Cu contains a coarser and more uniform grain structure even in the as-received condition, yielding resistivity values comparable to annealed Cu. While rolling-induced strain and defects may slightly influence resistivity, such variations fall within the margins of instrumental error and are difficult to discern. Moreover, the resistivity of the C/R Cu remains largely unchanged during subsequent annealing.

4. Discussion

The goal of this work was to demonstrate using a well characterized pure metal that PVD can produce thick, freestanding foils with bulk-like microstructures to enable combinatorial studies of structural materials. To demonstrate that combinatorial fabrication of thick, free-standing

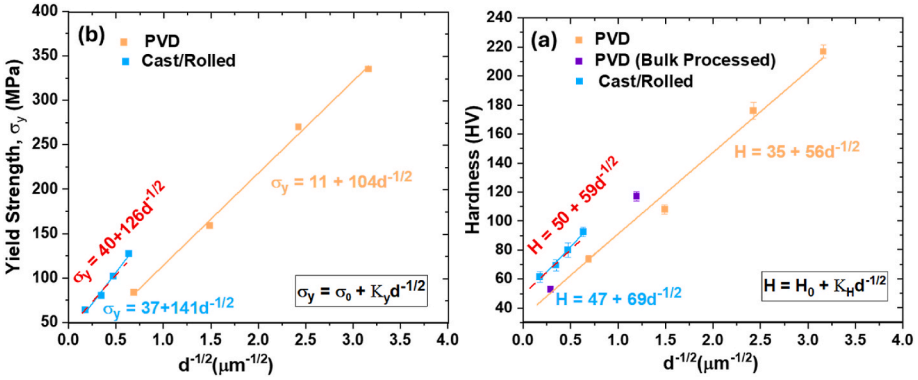


Fig. 9. Hall–Petch relationship for both foil types showing (a) yield strength and (b) hardness as a function of grain size. Data for PVD and C/R specimens are presented along with corresponding linear fits and associated equations. The red dashed line in each plot represents the fit obtained using only annealed C/R specimens. In (b), the two purple data points correspond to PVD foils after cold rolling and subsequent annealing. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

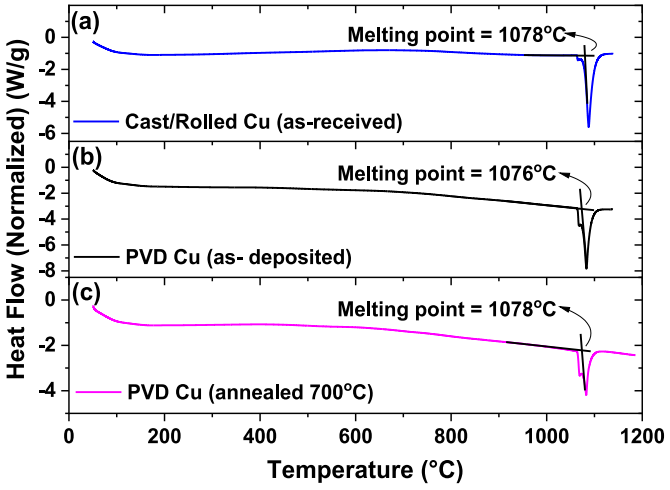


Fig. 10. Differential thermal analysis (DTA) scans of (a) C/R Cu, (b) as-deposited PVD Cu, and (c) 700 °C annealed PVD Cu, showing the onset of melting. The melting points for PVD Cu approach those of bulk Cu upon annealing, indicating bulk-like thermal behavior.

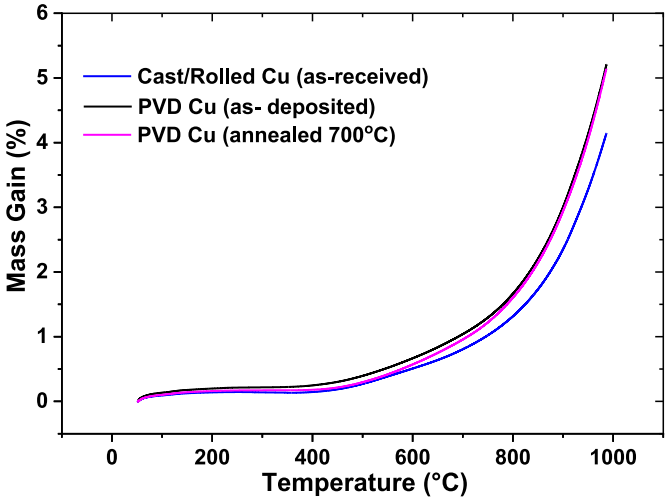


Fig. 11. Thermogravimetric analysis (TGA) showing mass gain as a function of temperature for as-received C/R Cu, as-deposited PVD Cu, and 700 °C annealed PVD Cu during heating in air up to 1000 °C.

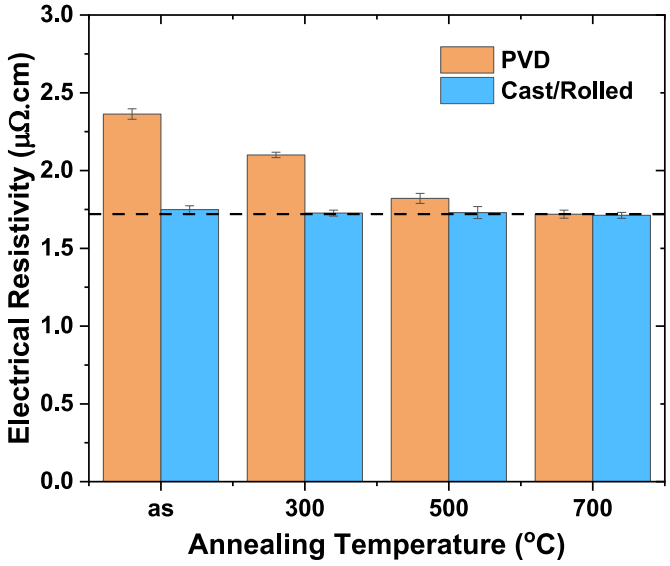


Fig. 12. Electrical resistivity of PVD and C/R Cu specimens in the as-deposited/as-received condition and after annealing at 300 °C, 500 °C, and 700 °C. The dashed line represents the theoretical resistivity of annealed bulk Cu (1.7241 μΩ.cm).

foils is possible, we present composition data in Fig. S2 for binary Cu–Ti foils revealing a Ti gradient from 0 to 8 at.% generated within a single deposition. Broader composition libraries will be examined in future work. To support these ongoing and future studies, here we explored whether bulk-like microstructures and properties can be obtained in thick PVD foils using pure Cu as a model material. We did so by comparing the microstructures and properties of PVD and commercial C/R Cu foils following thermal and mechanical processing.

4.1. *Impact of annealing on microstructures*

PVD and C/R Cu were annealed between 300 °C and 700 °C. Three features were tracked: grain size, texture, and Σ3 (twin) boundary fraction. In PVD Cu, grain size increased from ~100 nm to ~2.1 μm and the Σ3 fraction ranged from 38 % to 65.9 %. The increase in Σ3 fraction occurs due to the relatively low SFE of Cu that promotes the nucleation of annealing twins [44]. The as-deposited PVD specimens exhibit a strong {111} texture which is typical in PVD films driven by surface energy minimization and kinetic factors involved during deposition [57]. Upon annealing, the texture gradually evolved from a strong {111}

to a combination of {111} and {100}. {100} texture is favored at higher temperatures due to the nucleation of strain-free grains in the form of annealing twins. On the other hand, an increase in {111} texture is observed due to the abnormal grain growth of (111) grains to minimize the surface energy. The competition between surface energy-driven abnormal grain growth (favoring (111) grains) and strain energy minimization (favoring nucleation of new (100)-oriented twins) governs the texture evolution at elevated temperatures [57,58].

In C/R Cu, grain size increased from ~2.5 μm to ~32 μm; the Σ3 fraction rose from 44.5 % to 61.6 % at 500 °C and then decreased to 16.2 % at 700 °C. The sudden drop in Σ3 fraction suggests that many previously formed twins are consumed by highly mobile, high-angle grain boundaries during the coarsening process. Although Σ3 boundaries are typically low-energy and thermodynamically stable, they are susceptible to elimination in later stages of grain growth when boundary mobility is high and grain growth is no longer inhibited at elevated temperatures near 0.7–0.8 T_m [59]. The initial texture contained brass-type {011}⟨211⟩, copper-type {112}⟨111⟩, and S-type {123}⟨634⟩ components which is a well-known signature of deformed FCC metals [45]. The combination of texture remained even at higher temperatures; however, its intensity weakened on annealing.

A key difference between the two types of foils is the higher thermal stability of the PVD Cu. Ultrafine-grained Cu made by severe plastic deformation often coarsens rapidly, even at lower temperatures, due to high stored grain boundary energy [60,61]. In contrast, the PVD foils coarsened slowly from a nanocrystalline starting state and retain an average grain size of ~2 μm even after annealing at 700 °C. Two plausible contributors are: (i) a high density of nanotwins in PVD Cu (unresolvable by EBSD), which reduces boundary mobility [62,63]; and (ii) possible argon entrapment at boundaries that impedes migration [64]. These mechanisms were not directly measured and require further investigation.

4.2. Effect of microstructure on hardness and yield strength

As-deposited PVD Cu (nanocrystalline) shows a high yield strength (~335 MPa) and negligible ductility/strain hardening due to limited dislocation storage. Annealing at 300 °C lowers the yield strength to ~270 MPa and increases elongation to ~17 %; higher temperatures further reduce strength and increase ductility. When abnormal grain growth produces a bimodal grain structure, strain hardening increases modestly. In contrast, C/R Cu exhibits conventional coarse-grained behavior with a yield strength of ~128 MPa in the as-received state, ~65 MPa after 700 °C, and large uniform elongation (~45 %) with an initially higher hardening rate that decreases with strain. Overall, grain size and defect structure control the differences: PVD foils (fine grains) show lower work hardening, whereas C/R Cu (coarser grains) sustain classic dislocation-storage-driven hardening. Similarly, the hardness of PVD Cu drops from ~216 HV (as-deposited) to ~74 HV (700 °C) as grain size increases from ~0.1 to ~2.1 μm. C/R Cu decreases more modestly, ~92 to ~61 HV as grain size grows from ~2.5 to ~32 μm.

Both hardness and yield strength follow H–P trends within each foil type (Fig. 9). The extracted H–P constants are summarized in Table 2, alongside representative values reported in the literature. The values obtained, from both tension and hardness tests, fall well within the range of literature values. However, if we perform a direct comparison between the two foil types, PVD Cu shows lower H–P slopes and intercepts than C/R Cu, indicating a reduced grain-boundary strengthening contribution. From tensile-based H–P analysis (Fig. 9a), σ_0 is 40 MPa for C/R Cu which is higher than the measured value of 11 MPa for PVD Cu, but both fall within the values reported in literature for bulk Cu.

One factor that likely contributes to the difference in σ_0 is the fact that C/R Cu (mixed copper/brass-type; m ≈ 0.372–0.409) has a lower effective Schmid factor (m) than PVD Cu with dominant {111} texture (m ≈ 0. 408). This should raise the macroscopic yield for the same CRSS and thus the apparent H–P intercept. Along with σ_0, the H–P slope in C/R Cu (126 MPa $\mu m^{-1/2}$) is also higher than in PVD Cu (104 MPa $\mu m^{-1/2}$). This difference can be attributed to the mixed texture promoting a higher number of slip systems to be activated in the C/R Cu compared to a strong {111} texture in PVD Cu. This enables more non-coplanar slip systems across grains in C/R Cu which results in enhanced dislocation interactions, storage and pile-up, thereby increasing H–P slope.

Table 2
Summary of H_0,K_H,σ_0,K_y values obtained from Hall–Petch relationships derived in this study and reported in the literature, based on hardness and yield stress measurements.

	H_0 (HV)	K_H (HV·μm$^{-1/2}$)	σ_0 (MPa)	K_y (MPa·μm$^{-1/2}$)
PVD Cu (this study)	35 ± 6	56 ± 5	11 ± 8	104 ± 6
C/R Cu (this study)	50 ± 4	59 ± 10	40 ± 5	126 ± 9
Meric et al. [65]	13.4	52.2	–	–
Hakamada et al. [66]	80	67	–	–
Chokshi et al. [67]	52	60	–	–
Taha et al. [68]	25.2	31.6	–	–
Meyers et al. [69]	–	–	25.5	110
Hansen et al. [70]	–	–	20	140
Armstrong et al. [71]	–	–	25.5	111.5
Tian et al. [72]	–	–	9	111
Cordero et al. [73]	–	–	40	110
Merz et al. [74]	–	–	6.39	118
Thompson et al. [75]	–	–	3.4	140

In the hardness-based H–P analysis (Fig. 9b), the effect of texture diminishes as the multiaxial stress-state and substantial localized plastic deformation (8–10 %) under the indenter activates many slip systems and reduces texture sensitivity [56]. Thus, as shown in Table 2, the differences between H–P intercepts and slopes for PVD and C/R Cu hardness diminish compared to the tensile data. Further still, cold rolling and annealing the PVD foils drives the hardness values towards the C/R Cu H-P trend as shown by the purple data points in Fig. 9b. This suggests that mechanical properties will be very similar as the microstructures approach bulk-like Cu.

4.3. Effect of microstructure on physical properties and oxidation behavior

The apparent melting point for C/R Cu matched well with annealed (coarser-grained) PVD Cu as shown in Fig. 10. The as-deposited PVD Cu shows a small (~2 °C) lower onset, consistent with finer grains and higher boundary volume fraction [76] but reverts to the expected value for pure Cu with annealing. Regarding oxidation behavior, after normalizing for specimen thickness, the overall oxidation-induced mass gain was similar for PVD and C/R Cu. However, differences in the progression of oxidation arise due to microstructure. The earlier oxidation onset in the as-deposited PVD Cu is consistent with its much finer initial grain size (~100 nm). Smaller grains increase the grain-boundary fraction and grain boundaries are more prone to oxidation and act as fast oxygen diffusion paths, which increases mass gain [77]. With heating, substantial grain coarsening reduces boundary area, and the as-deposited PVD foil approaches the microstructure of the annealed PVD foil. This explains the convergence of their mass-gain curves near ~750 °C.

The higher resistivity of as-deposited PVD Cu is explained by grain-boundary scattering in its ultrafine-grained microstructure [40]. Prior studies have shown that PVD Cu with grain sizes on the order of 15 nm can exhibit resistivity values up to ten times higher than those of annealed bulk Cu [39]. As the electron mean free path becomes comparable to the grain size, enhanced scattering leads to a pronounced increase in resistivity, especially at nanometer-scale dimensions [78]. In this study, the as-deposited PVD specimen has an average grain size of ~200 nm, resulting in its resistivity being approximately 1.4 times

greater than bulk Cu. Upon annealing at intermediate temperatures, a gradual reduction in resistivity is observed, although the values remain elevated due to the retention of relatively fine grains. When annealed at 700 °C, the microstructure becomes more homogeneous with significantly larger grains, causing the resistivity to approach that of a fully annealed Cu (1.7241 μΩ.cm). C/R Cu starts with coarser, more uniform grains and fewer defects, so its resistivity is near the annealed value from the outset and remains essentially unchanged with annealing, consistent with minimal microstructural evolution that could affect electrical properties.

4.4. Overall comparison between PVD and C/R Cu

PVD and C/R Cu foils exhibit different initial microstructural features, including texture, dislocation density, and grain size, yet their evolution with annealing and processing converges toward similar bulk-like behavior. Both systems follow conventional H–P scaling, with differences in intercepts and slopes arising primarily due to variations in texture. Importantly, the strengthening response of PVD Cu progressively approaches that of C/R Cu as its microstructure starts to become comparable through cold rolling and annealing. Similarly, initial differences in resistivity, oxidation behavior, and apparent melting temperature in PVD Cu diminish upon annealing, ultimately converging to values characteristic of bulk Cu. These parallels confirm that by thermomechanical processing thick, free-standing PVD foils one can replicate the essential mechanical, electrical, and thermal responses of C/R Cu, thereby demonstrating that PVD foils offer an effective basis for performing combinatorial studies on structural alloys.

5. Conclusions

This study demonstrates that freestanding Cu foils fabricated via magnetron sputtering can achieve bulk-like microstructures and behavior that closely resembles conventionally processed Cu. With thicknesses approaching 200 μm, PVD foils exhibit grain growth, texture evolution, annealing twin formation, and H–P strengthening trends consistent with those observed in C/R foils after appropriate thermal and mechanical treatments. The PVD foils also withstand substantial plastic deformation during cold rolling and subsequently undergo recrystallization, producing a grain structure and texture similar to bulk-like Cu. Thermal and physical property measurements further support this equivalence: melting points and oxidation kinetics were comparable between the two material classes, while electrical resistivity of the PVD foils converged toward bulk values following grain coarsening at elevated annealing temperatures.

While the microstructures of PVD and C/R Cu inevitably differ, this comparison provides a practical reference point and establishes that following thermal and mechanical processing thick PVD foils can achieve mechanical and physical properties within the same performance envelope as conventionally processed Cu. This capability enables the integration of high-throughput combinatorial PVD synthesis with bulk-scale mechanical and physical property characterization, creating a powerful framework to accelerate the discovery and qualification of advanced structural alloys.

CRediT authorship contribution statement

Rohit Berlia: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chetan Singh:** Writing – review & editing, Validation, Software, Investigation, Formal analysis. **Zhaosen Qu:** Software, Methodology, Investigation, Data curation. **Rayna T. Mehta:** Writing – review & editing, Software, Methodology, Investigation, Data curation. **Timothy P. Weihs:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.msea.2025.149163.

Data availability

The data supporting the findings of this study are included within the article and its supplementary materials. Additional data is available from the corresponding author upon request.

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